

lab_09

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```
library(readr)
```

Warning: package 'readr' was built under R version 4.5.2

```
library(bio3d)  
library(ggplot2)  
library(tidyr)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v forcats 1.0.1      v stringr 1.5.2
v lubridate 1.9.4    v tibble 3.3.0
v purrr      1.1.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
pdb_stats <- as.data.frame(read_csv("pdb_stats.csv"))
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
Delimiter: ","
chr (1): Molecular_Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X_ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
cols_to_pivot <- colnames(pdb_stats)[2:8]
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

0.8143231

Q2: What proportion of structures in the PDB are protein?

0.9791355

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

513

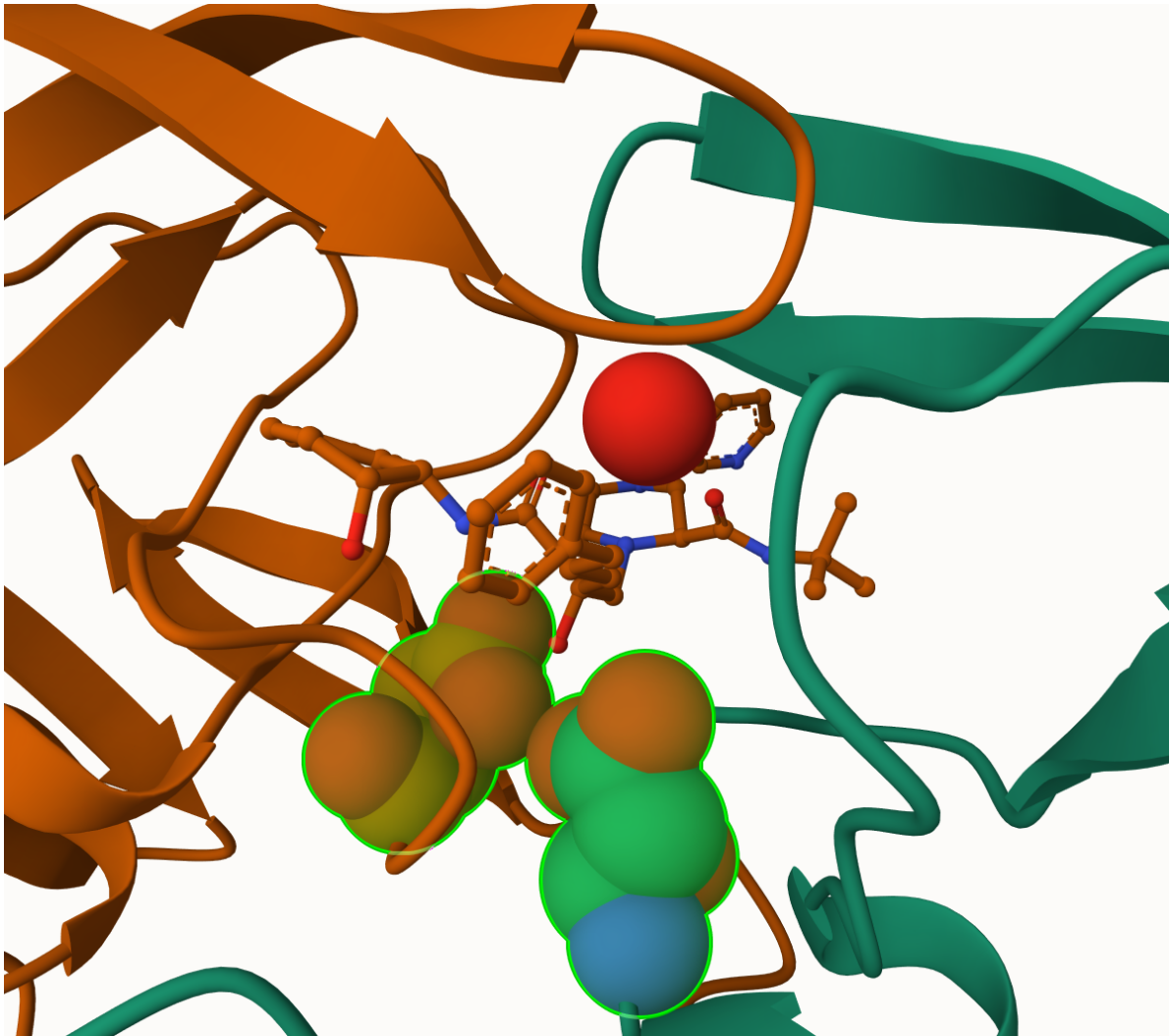


Figure 1: 1HSG

```
pdb_long <- pivot_longer(pdb_stats, cols=cols_to_pivot, names_to="category", values_to = "co
```

Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.

i Please use `all\_of()` or `any\_of()` instead.

Was:

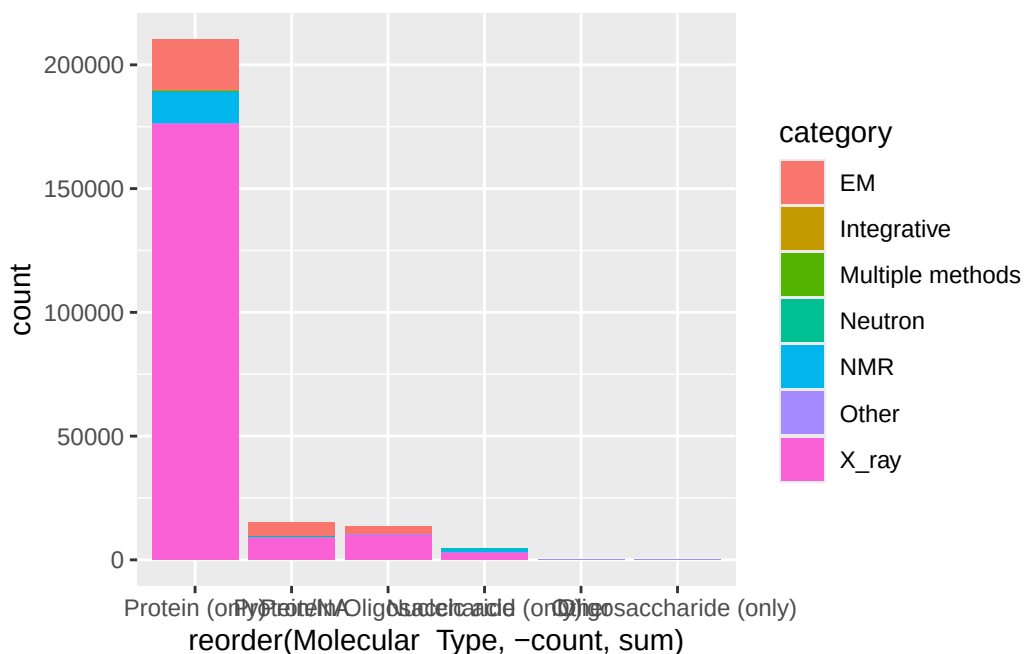
```
data %>% select(cols_to_pivot)
```

Now:

```
data %>% select(all_of(cols_to_pivot))
```

See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.

```
ggplot(pdb_long) +  
  aes(x=reorder(Molecular_Type, -count, sum), y=count, fill=category) +  
  geom_bar(stat="identity", position="stack")
```



Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

To reduce visual clutter. Molstar defaults to showing the oxygen atom only.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

residue 308

```
hsg <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hsg
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

Nucleic atoms

Q9: How many protein chains are in this structure?

2

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

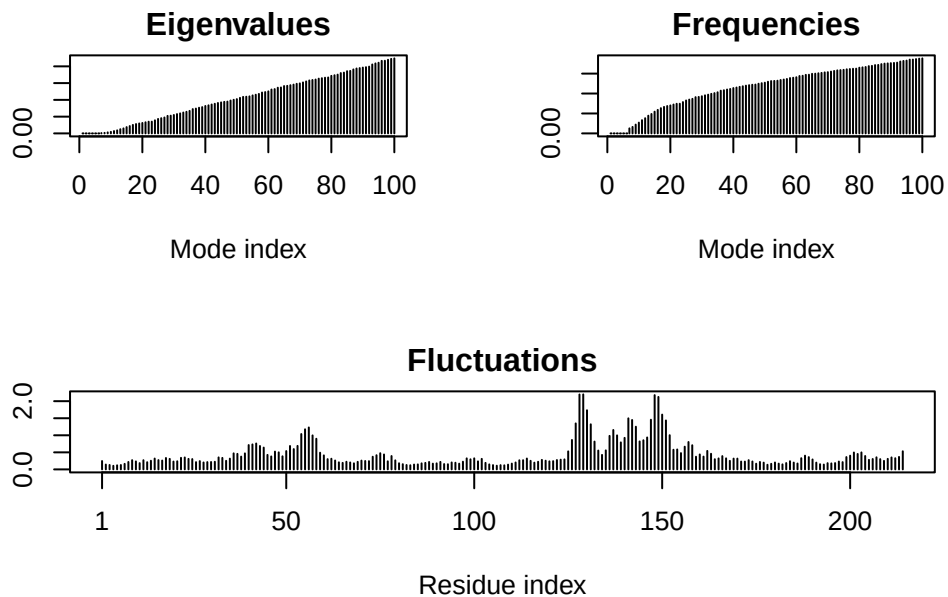
PDB has ALT records, taking A only, rm.alt=TRUE

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.06 seconds.
```

```
Diagonalizing Hessian... Done in 0.46 seconds.
```

```
mktrj(m, file="adk_m7.pdb")
plot(m)
```



Q10. Which of the packages above is found only on BioConductor and not CRAN?

BiocManager

Q11. Which of the above packages is not found on BioConductor or CRAN?

“Grantlab/bio3d-view”

True or False? Functions from the devtools package can be used to install packages from GitHub and BitBucket?

True

```
aa <- get.seq("lake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

     121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214